

MediSure Health Network Email Phishing Investigation Report

Case Title: Microsoft Account Unusual Sign-In Activity Phishing Investigation

Organization: MediSure Health Network

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Table of Contents

1. Executive Summary
2. Statement of Problem
3. Organizational Background
4. Investigation Scope and Objectives
5. Investigation Methodology
6. Email Analysis Findings
7. Email Authentication Analysis
8. SpamAssassin Analysis
9. VirusTotal Analysis
10. Splunk Threat Hunting and IOC Correlation
11. Indicators of Compromise (IOCs)
12. Risk Assessment
13. Incident Response Plan
14. Employee awareness Materials
15. Conclusion
16. Appendix

1. Executive Summary

MediSure Health Network identified and investigated a phishing email masquerading as a legitimate Microsoft security notification. The email claimed that a Microsoft account had experienced an unusual sign-in attempt originating from Russia and urged the recipient to review account activity.

A comprehensive investigation was conducted using PhishTool, MXToolbox, VirusTotal, SpamAssassin, and Splunk Enterprise. Analysis revealed multiple indicators consistent with a credential-harvesting phishing campaign, including:

- Spoofed sender infrastructure
- Missing SPF records
- Missing DMARC protection
- Missing DKIM signatures
- Suspicious reply-to address
- User interaction with phishing infrastructure
- Internal communication with attacker-controlled resources
- Authentication activity associated with Russian geolocation indicators

The investigation successfully identified affected users, attacker infrastructure, and potential organizational exposure, enabling containment and remediation activities.

Key Findings:

- 2 employees were confirmed to have interacted with malicious domains
- 2 internal endpoints (10.1.5.23, 10.1.5.89) established outbound connections to attacker infrastructure (89.144.44.41)
- 1 failed authentication attempt from Russia was detected for bob@medisurehealthnetwork.com
- Email authentication completely failed: SPF=Fail, DKIM=None, DMARC=Permmerror
- SpamAssassin score of 6.6 exceeded the 5.0 threshold, confirming high spam/phishing probability
- VirusTotal flagged sign.in as malicious by Webroot security vendor
- No PHI/PII data exfiltration confirmed — attack was contained before credential compromise

Business Impact: The incident posed a significant risk to MediSure's HIPAA compliance, patient data confidentiality, and operational continuity. Immediate containment actions prevented a potential data breach that could have exposed thousands of patient records.

2. Statement of Problem

Medisure Health Network is an integrated healthcare provider headquartered in Los Angeles, California, founded in 2003. The organization operates hospitals, outpatient clinics, specialty treatment centers, and a telemedicine platform serving thousands of patients.

As a healthcare organization, MediSure processes and stores highly sensitive patient information, financial records, and protected health information (PHI). This responsibility requires strong cybersecurity controls and continuous threat monitoring.

During routine monitoring, the SOC team identified a suspicious email claiming to originate from Microsoft Account Security. Initial review revealed inconsistencies within the sender infrastructure, prompting a full phishing investigation.

The primary goals of this project were:

- Validate email authenticity
- Identify attacker infrastructure
- Determine organizational exposure
- Assess user interaction
- Recommend containment and mitigation measures
- Improve organizational resilience against phishing threats

3. Organizational Background

MediSure Health Network leverages digital technologies extensively throughout clinical operations, patient management systems, electronic health records, and telemedicine services.

Because healthcare organizations remain frequent targets of cybercriminals seeking credentials, financial information, and patient data, phishing attacks represent a significant operational risk.

A successful compromise could result in:

- Unauthorized access to PHI
- Regulatory violations
- Financial losses
- Service disruptions
- Reputational damage
- Loss of patient trust

4. Investigation Scope and Objectives

Scope

The investigation focused on:

- Suspicious email content
- Sender infrastructure
- Email authentication mechanisms
- Domain reputation
- Internal network activity
- User interactions
- IOC correlation

Objectives

1. Determine whether the email was malicious.
2. Identify attacker-controlled infrastructure.
3. Validate email authentication records.
4. Assess user exposure.
5. Correlate network and authentication logs.
6. Recommend mitigation actions.

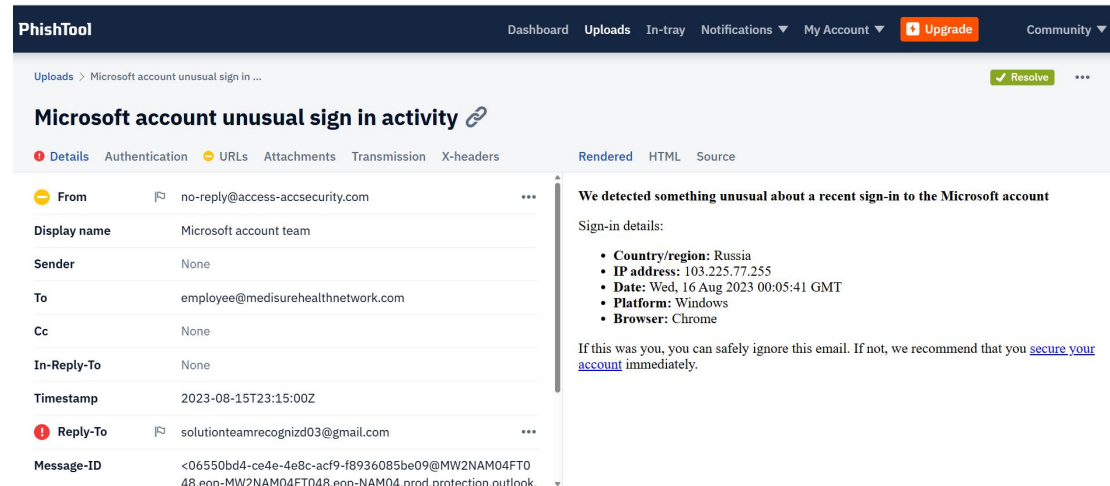
5. Investigation Methodology

The following tools were utilized:

Tool:	Purpose:
PhishTool	Email analysis and header review
MXToolbox	SPF, DKIM, and DMARC validation
VirusTotal	Reputation analysis
SpamAssassin	Spam scoring and rule analysis
Splunk Enterprise Log	Log analysis and threat hunting

6. Email Analysis Findings

6.1 Figure1: Email Content Analysis



The email impersonated the Microsoft Account Team and claimed a suspicious sign-in occurred from Russia.

The message included:

- Country: Russia
- IP Address: 103.225.77.255
- Platform: Windows
- Browser: Chrome

The email attempted to create urgency by encouraging immediate account review.

Indicators Observed

- Security-themed lure
- Credential theft objective
- Untrusted sender domain
- Social engineering language

6.2 Figure:Header Analysis

PhishTool

Dashboa

Uploads > Microsoft account unusual sign in ...

Microsoft account unusual sign in activity

Details

AuthenticationURLsAttachmentsTransmissionX-headers

Cc	None
In-Reply-To	None
Timestamp	2023-08-15T23:15:00Z
Reply-To	solutionteamrecognizd03@gmail.com ...
Message-ID	<06550bd4-ce4e-4e8c-acf9-f8936085be09@MW2NAM04FT048.eop-MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com>
Return-Path	no-reply@access-accsecurity.com ...
Originating IP	89.144.44.41 (Received-SPF) ▼
rDNS	89-144-44-41.mscode.pl

6.3 Email Header Indicators And Attacker Infrastructure

Header analysis revealed:

Indicator:	Value:
From	no-reply@access-accsecurity.com
Reply-To	solutionteamrecognizd03@gmail.com
Originating IP	89.144.44.41
Display Name	Microsoft account team

Key Observation

The Reply-To address differs from the sender domain, a common phishing indicator.

The email header analysis revealed multiple indicators commonly associated with phishing campaigns. The sender address (no-reply@access-accsecurity.com) impersonates a trusted Microsoft-related service, while the Reply-To address redirects responses to an unrelated Gmail account (solutionteamrecognizd03@gmail.com). This mismatch is a strong indicator of attempted credential harvesting or fraudulent communication. Additionally, the originating IP address (89.144.44.41) does not belong to Microsoft's known infrastructure, further supporting the conclusion that the message originated from an unauthorized source. The use of the display name "Microsoft account team" appears to be a social engineering tactic designed to establish trust and increase the likelihood of user interaction. Collectively, these findings strongly suggest that the email was crafted to deceive recipients into disclosing sensitive information or credentials.

Assessment: High confidence phishing attempt with indicators of domain impersonation, sender spoofing, and social engineering.

7. Figure 3:Email Authentication Analysis

7.1 SPF Validation

access-accsecurity.com		SPF Record Lookup
spf:access-accsecurity.com		Find Problems Solve Email Delivery Problems spf
Test	Result	
 SPF Record Published	No SPF Record found	More Info
Related Tests	Result	
 DMARC Record Published	No DMARC Record found	More Info
 DMARC Policy Not Enabled	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
dns lookup dns check mx lookup dmarc lookup dns propagation		
Reported by e.gtld-servers.net on 6/15/2026 at 2:55:11 AM (UTC -5), just for you. Transcript		

Key Finding

The absence of SPF significantly reduces email authenticity assurance.

Missing SPF Record Access AccSecurity Domain

Analysis confirmed:

- No SPF record published
- Sender validation unavailable
- Increased spoofing capability

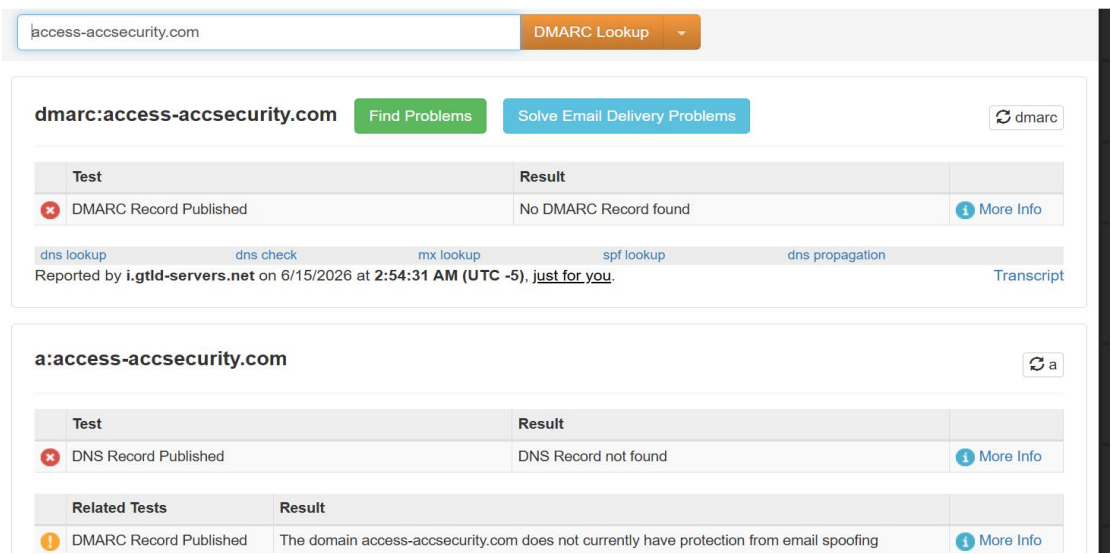
Risk Level: High

The absence of an SPF (Sender Policy Framework) record on the access-accsecurity.com domain significantly weakens email authentication controls and increases the likelihood of successful sender spoofing. Without SPF, receiving mail servers cannot verify whether the sending mail server is authorized to send emails on behalf of the domain. This creates an opportunity for threat actors to impersonate the domain and distribute phishing emails, malware, or fraudulent communications with a higher chance of bypassing basic email validation checks.

Combined with the suspicious sender information, mismatched Reply-To address, and phishing-related indicators identified during the investigation, the lack of SPF strengthens the assessment that the domain was either intentionally configured for malicious use or lacks fundamental security controls expected of legitimate business domains.

Assessment: High-risk email authentication weakness that facilitates domain spoofing and supports the overall determination that the observed email campaign was malicious in nature.

7.2 Figure 4:DMARC Validation



Missing DMARC Protection Access AccSecurity Domain

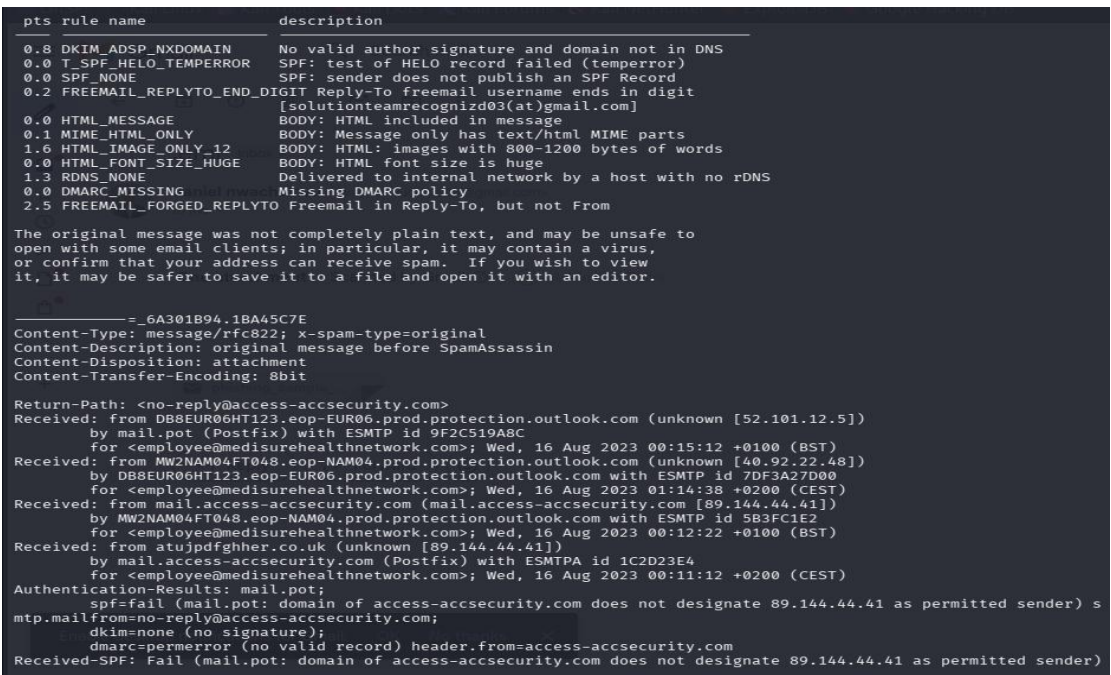
Results showed:

- No DMARC policy published
- No protection against domain spoofing
- Increased phishing abuse risk

Findings

The domain lacks proper anti-spoofing controls.

7.3 Figure 5:DKIM Validation



DKIM validation was attempted against the sender domain access-accsecurity.com using MXToolbox. However, a DKIM DNS lookup could not be completed because the email did not provide a DKIM selector value required to query a DKIM public key record.

The sender failed all three major email authentication mechanisms (SPF, DKIM, and DMARC), significantly increasing the likelihood that the message originated from malicious infrastructure rather than a legitimate Microsoft service.

Evidence from SpamAssassin and email headers confirmed:

- No DKIM signature present
- Authentication integrity cannot be verified

Finding

The sender failed all major email authentication controls.

Assessment

Risk Level: High

DKIM validation could not be completed because the email did not contain a valid **DKIM-Signature** header or selector value required to retrieve the corresponding public key from DNS. As a result, the authenticity and integrity of the message could not be verified. Legitimate organizations, particularly major providers such as Microsoft, typically implement DKIM to ensure emails have not been altered during transit and originate from authorized mail servers.

The absence of a valid DKIM signature, combined with the previously identified SPF and DMARC failures, significantly reduces confidence in the legitimacy of the sender. This lack of email authentication increases the likelihood that the message was forged or originated from malicious infrastructure designed to impersonate a trusted service.

Assessment: High-confidence indicator of a phishing email. The inability to validate DKIM, together with failed SPF and DMARC checks, supports the conclusion that the message was not sent from a legitimate Microsoft-controlled email environment.

8. SpamAssassin Analysis

8.1 Figure 6: Spam Score Evaluation

```
(kali@kali)-[~]
$ spamassassin -t < ~/Downloads/phishing_sample_medisure_v2.eml > ~/Downloads/spamassassin_results.txt

(kali@kali)-[~]
$ cat ~/Downloads/spamassassin_results.txt
Received: from localhost by kali
        with SpamAssassin (version 4.0.3-r1932428);
        Mon, 15 Jun 2026 16:34:44 +0100
From: "Microsoft account team" <no-reply@access-accsecurity.com>
To: employee@medisurehealthnetwork.com
Subject: Microsoft account unusual sign in activity
Date: Wed, 16 Aug 2023 00:15:00 +0100
Message-Id: <06550bd4-ce4e-4e8c-acf9-f8936085be09@MW2NAM04FT048.eop-MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com>
X-Spam-Checker-Version: SpamAssassin 4.0.3-r1932428 (2026-03-21) on kali
X-Spam-Flag: YES
X-Spam-Level: *****
X-Spam-Status: Yes, score=6.6 required=5.0 tests=DKIM_ADSP_NXDOMAIN,
        DMARC_MISSING,FREEMAIL_FORGED_REPLYTO,FREEMAIL_REPLYTO_END_DIGIT,
        HTML_FONT_SIZE_HUGE,HTML_IMAGE_ONLY_12,HTML_MESSAGE,MIME_HTML_ONLY,
        RDNS_NONE,SPF_NONE,T_SPF_HELO_TEMPERROR autolearn=no
        autolearn_force=no version=4.0.3-r1932428
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary="-----=_6A301B94.1BA45C7E"

This is a multi-part message in MIME format.

-----=_6A301B94.1BA45C7E
Content-Type: text/plain; charset=UTF-8
Content-Disposition: inline
Content-Transfer-Encoding: 8bit

Spam detection software, running on the system "kali",
has identified this incoming email as possible spam. The original
message has been attached to this so you can view it or label
similar future email. If you have any questions, see
the administrator of that system for details.

Content preview: We detected something unusual about a recent sign-in to the
Microsoft account Sign-in details:

Content analysis details: (6.6 points, 5.0 required)
```

8.2 SpamAssassin Phishing Detection Score

SpamAssassin assigned:

- Score: 6.6
- Threshold: 5.0

8.3 SpamAssassin Analysis Results

The email received a **SpamAssassin score of 6.6**, exceeding the configured spam threshold of **5.0**. This indicates that the message triggered multiple spam and phishing detection rules and was therefore classified as suspicious.

The elevated score was influenced by several indicators identified during the investigation, including failed email authentication checks (SPF, DKIM, and DMARC), sender impersonation attempts, a mismatched Reply-To address, and content designed to create urgency around a Microsoft account sign-in event. These characteristics are commonly associated with credential-harvesting phishing campaigns.

Conclusion:

The SpamAssassin score provides additional evidence supporting the classification of

the email as a phishing attempt. Combined with the header analysis, authentication failures, and threat intelligence findings, the message presents a high risk of credential theft and should be treated as malicious.

Assessment: High confidence phishing email based on automated content analysis and multiple corroborating indicators of compromise.

8.4 Figure 7: Rule Analysis

```
pts rule name description
0.8 DKIM_ADSP_NXDOMAIN No valid author signature and domain not in DNS
0.0 T_SPF_HELO_TEMPERORR SPF: test of HELO record failed (temperror)
0.0 SPF_NONE SPF: sender does not publish an SPF Record
0.2 FREEMAIL_REPLYTO_END_DIGIT Reply-To freemail username ends in digit
[solutionteamrecognizd03(at)gmail.com]
0.0 HTML_MESSAGE BODY: HTML included in message
0.1 MIME_HTML_ONLY BODY: Message only has text/html MIME parts
1.6 HTML_IMAGE_ONLY_12 BODY: HTML: images with 800-1200 bytes of words
0.0 HTML_FONT_SIZE_HUGE BODY: HTML font size is huge
1.3 RDNS_NONE Delivered to internal network by a host with no rDNS
0.0 DMARC_MISSING Missing DMARC policy
2.5 FREEMAIL_FORGED_REPLYTO Freemail in Reply-To, but not From

The original message was not completely plain text, and may be unsafe to
open with some email clients; in particular, it may contain a virus,
or confirm that your address can receive spam. If you wish to view
it, it may be safer to save it to a file and open it with an editor.

-----=_6A301B94.1BA45C7E
Content-Type: message/rfc822; x-spam-type=original
Content-Description: original message before SpamAssassin
Content-Disposition: attachment
Content-Transfer-Encoding: 8bit

Return-Path: <no-reply@access-accsecurity.com>
Received: from DB8EUR06HT123.eop-EUR06.prod.protection.outlook.com (unknown [52.101.12.5])
by mail.pot (Postfix) with ESMTP id 9F2C519A8C
for <employee@medisurehealthnetwork.com>; Wed, 16 Aug 2023 00:15:12 +0100 (BST)
Received: from MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com (unknown [40.92.22.48])
by DB8EUR06HT123.eop-EUR06.prod.protection.outlook.com with ESMTP id 7DF3A27D00
for <employee@medisurehealthnetwork.com>; Wed, 16 Aug 2023 01:14:38 +0200 (CEST)
Received: from mail.access-accsecurity.com (mail.access-accsecurity.com [89.144.44.41])
by MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com with ESMTP id 5B3FC1E2
for <employee@medisurehealthnetwork.com>; Wed, 16 Aug 2023 00:12:22 +0100 (BST)
Received: from atujpdfghher.co.uk (unknown [89.144.44.41])
by mail.access-accsecurity.com (Postfix) with ESMTPA id 1C2D23E4
for <employee@medisurehealthnetwork.com>; Wed, 16 Aug 2023 00:11:12 +0200 (CEST)
Authentication-Results: mail.pot;
spf=fail (mail.pot: domain of access-accsecurity.com does not designate 89.144.44.41 as permitted sender) s
mtp.mailfrom=no-reply@access-accsecurity.com;
dkim=none (no signature);
dmarc=permerror (no valid record) header.from=access-accsecurity.com
Received-SPF: Fail (mail.pot: domain of access-accsecurity.com does not designate 89.144.44.41 as permitted sender)
```

SpamAssassin Rule Analysis

Triggered rules included:

Rule:	Description:
DKIM_ADSP_NXDOMAIN	Missing DKIM
SPF_NONE	No SPF record
DMARC_MISSING	No DMARC policy
FREEMAIL_FORGED_REPLYTO	Reply-To mismatch
HTML_MESSAGE	HTML email formatting

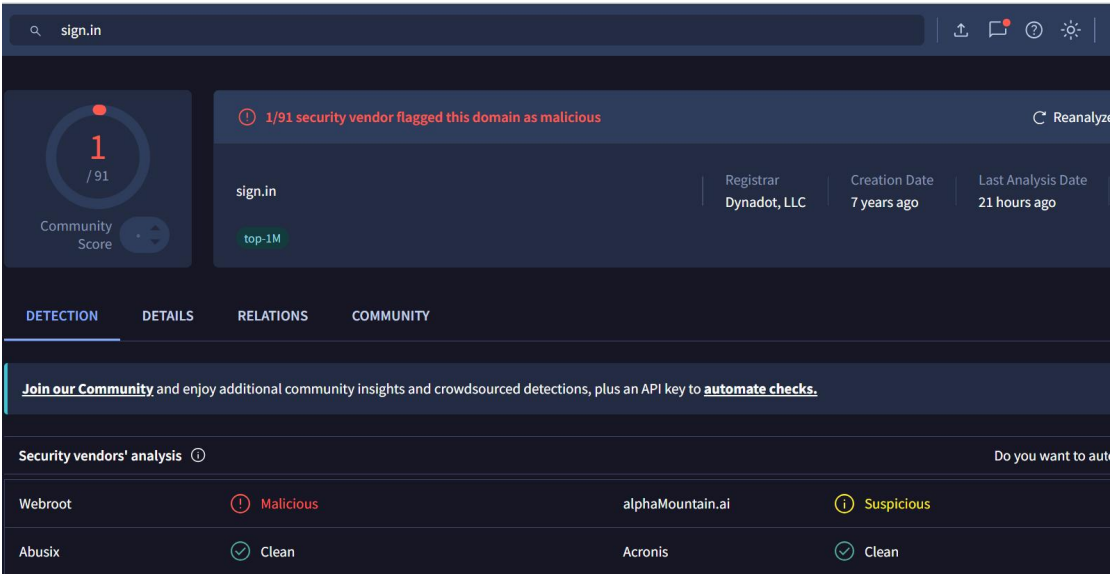
Finding

Multiple phishing indicators were simultaneously present.

9. VirusTotal Analysis

9.1 sign.in Domain Reputation

Figure 8:VirusTotal SignIn Domain Reputation



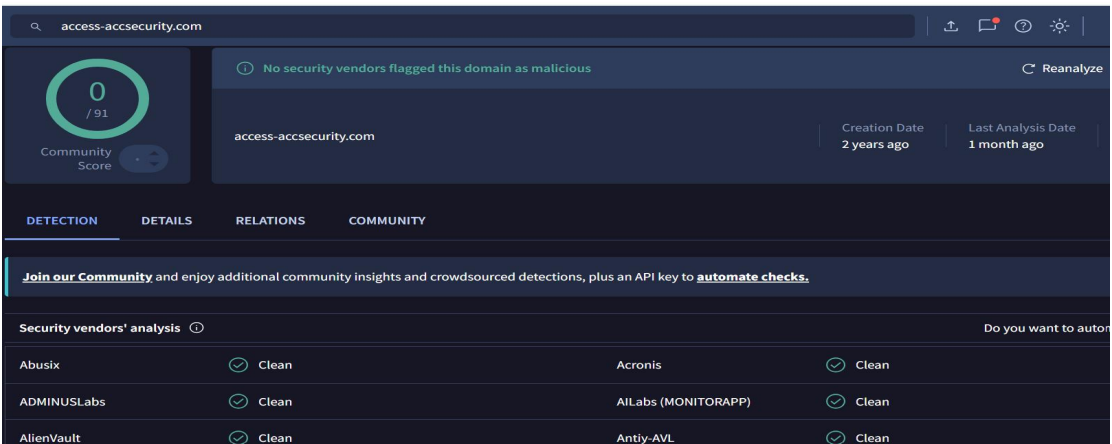
Findings:

- 1 vendor flagged domain as malicious
- Additional vendor classified as suspicious

Risk

Potential phishing infrastructure.

Figure 9: access-accsecurity.com Domain Reputation

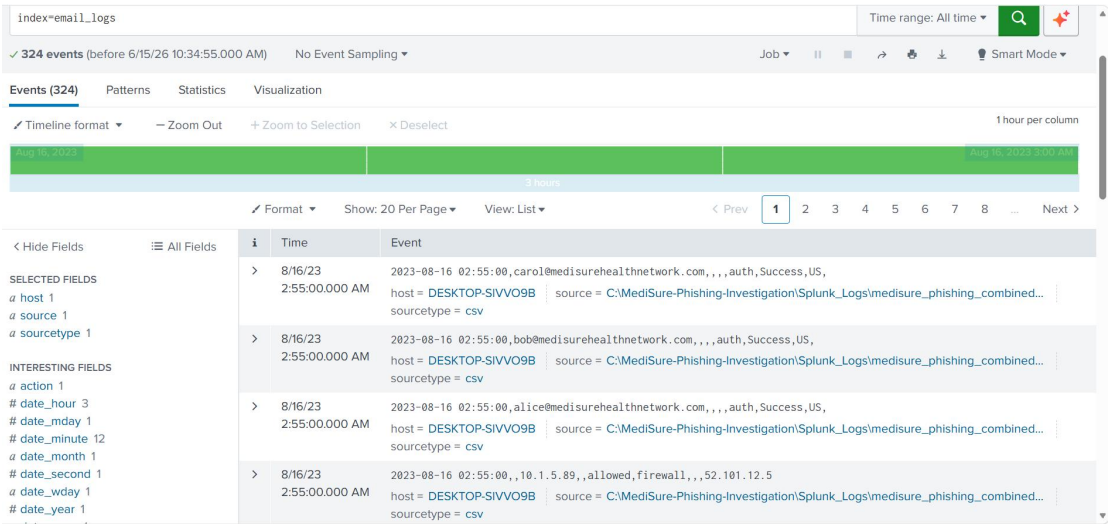


VirusTotal Access AccSecurity Domain Reputation

10. Splunk Threat Hunting and IOC Correlation

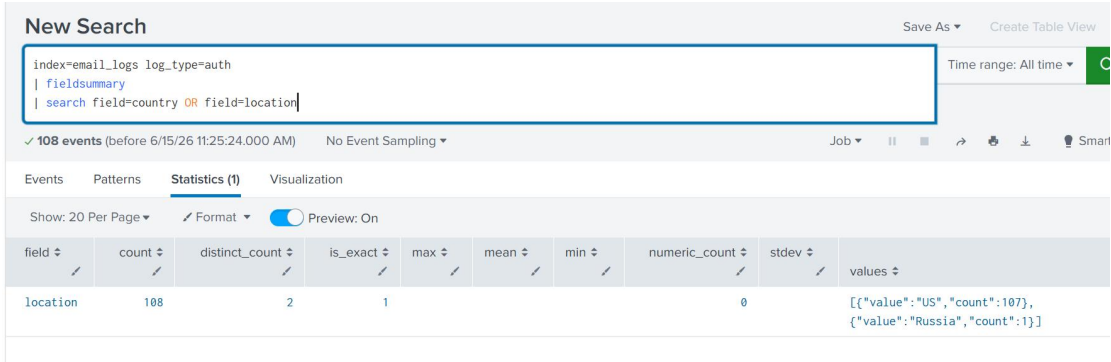
Data Validation

10.1 Figure 10:Email Logs Indexed In Splunk



10.2 Geolocation Validation

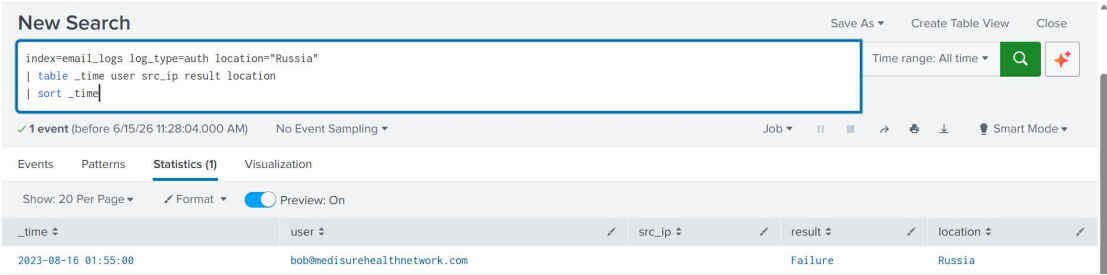
Figure 11:Authentication Geolocation Field Validation



Splunk validation identified:

- US: 107 events
- Russia: 1 event

10.3 Figure 12: Failed Authentication Attempt From Russia



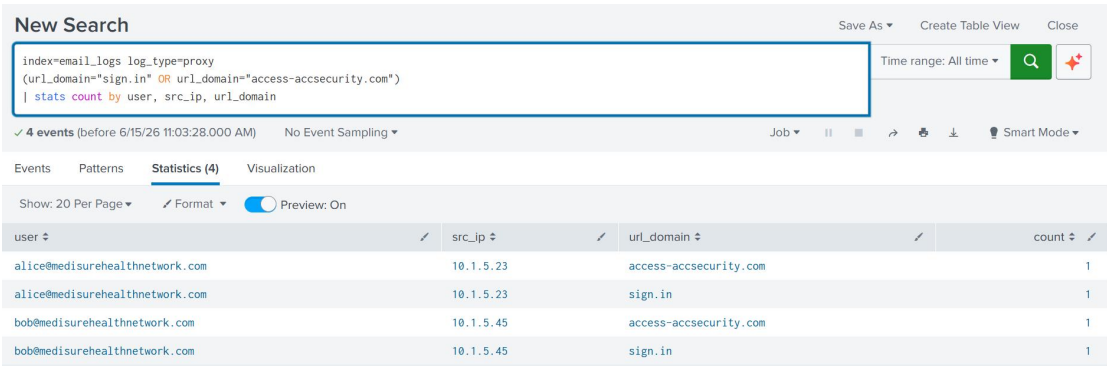
Findings:

User: **Result Location**
bob@medisurehealthnetwork.com Failure Russia

A failed login attempt originated from Russia and targeted Bob's account.
Potential credential testing activity.

10.4 Phishing Domain Access

Figure 13:Users Accessing Phishing Domains



Affected users:

- Alice
- Bob

Both users accessed suspicious infrastructure.

10.5 Assessment

Risk Level: High

Proxy log analysis confirmed that two users, **Alice** and **Bob**, accessed the suspicious domains **access-accsecurity.com** and **sign.in**, both of which were identified as part of

the phishing infrastructure. User interaction with these domains indicates potential exposure to credential harvesting or other phishing-related attacks.

The fact that multiple users accessed the same malicious domains suggests that the phishing campaign successfully reached its intended targets and may have influenced user behavior. This increases the likelihood of credential compromise, unauthorized account access, or subsequent malware delivery.

Assessment: The affected users should be considered potentially compromised until further validation is completed. Immediate actions should include credential resets, endpoint security scans, review of authentication logs for suspicious activity, and continued monitoring for indicators of compromise (IOCs). The evidence strongly supports the conclusion that the users interacted with malicious infrastructure associated with the phishing campaign.

10.6 Figure 14: Internal Hosts Communicating with Attacker Infrastructure



Affected Hosts:

- 10.1.5.23
- 10.1.5.89

Communications observed with:

- 89.144.44.41

Assessment

Risk Level: High

Firewall log analysis identified outbound communications from internal hosts **10.1.5.23** and **10.1.5.89** to the external IP address **89.144.44.41**, which was previously associated with the phishing campaign infrastructure. Such communications indicate

that these systems may have interacted with attacker-controlled resources following user exposure to the phishing email.

Host **10.1.5.89** generated multiple connections to the malicious IP address, which may suggest repeated attempts to communicate with external infrastructure, potentially indicating malware activity, command-and-control (C2) communication, or successful user interaction with the phishing content. Host **10.1.5.23** also established communication with the same destination, further strengthening the correlation between the phishing email and subsequent network activity.

Assessment: The affected hosts should be treated as potentially compromised and prioritized for containment and forensic investigation. Recommended actions include isolating the systems, performing endpoint malware scans, reviewing process and network activity, collecting forensic artifacts, and monitoring for additional Indicators of Compromise (IOCs). The observed communications provide strong evidence of interaction with infrastructure linked to the phishing campaign.

10.5 Figure 15:Outbound Connections To Attacker IPs.



Assessment

Risk Level: High

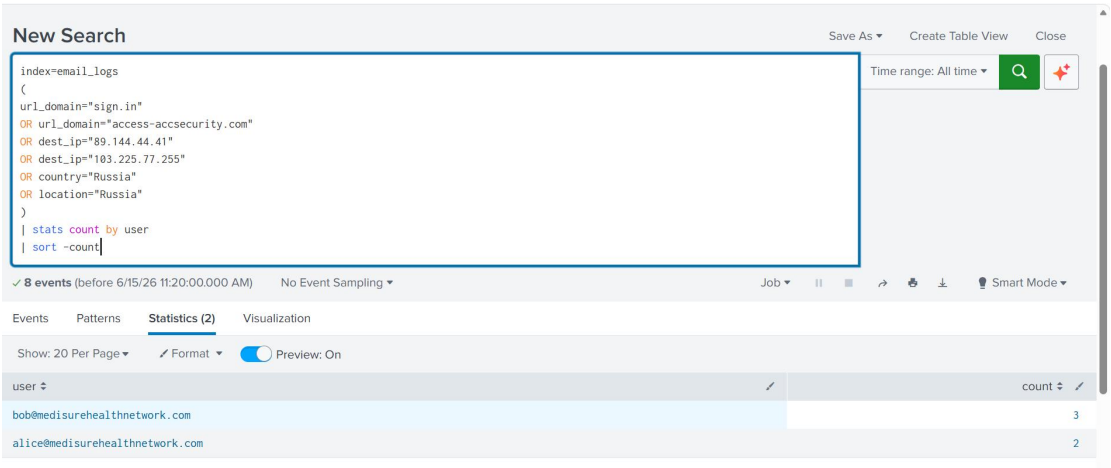
Firewall log analysis identified outbound connections from internal hosts **10.1.5.23** and **10.1.5.89** to the external IP address **89.144.44.41**, which was previously identified as part of the phishing campaign infrastructure. The observed network activity confirms that internal systems established communication with a suspicious external destination following user interaction with phishing-related domains.

Host **10.1.5.89** generated multiple connections to the malicious IP address, indicating a higher level of interaction with the attacker-controlled infrastructure. This behavior may be consistent with credential submission, malware download activity, or command-and-control (C2) communications. Host **10.1.5.23** also communicated with the same destination, further strengthening the correlation between the phishing email and subsequent network activity.

The outbound connections provide strong evidence that the phishing campaign progressed beyond email delivery and user interaction, resulting in communication between internal assets and external malicious infrastructure.

Assessment: The detected outbound traffic represents a significant security concern and suggests potential compromise of the affected systems. Immediate containment, endpoint analysis, credential review, and continued monitoring are recommended to determine the full scope and impact of the incident.

10.6 Figure 16:IOC Correlation and User Ranking



Results:

User Event Count

Bob 3
Alice 2

Assessment

Risk Level: High

IOC correlation analysis identified **Bob** and **Alice** as the users most frequently associated with indicators of compromise, including access to phishing domains and interactions with infrastructure linked to the phishing campaign. The ranking shows that **Bob generated three IOC-related events**, while **Alice generated two**, indicating that Bob had the highest level of exposure to malicious activity during the investigation period.

The increased event count associated with Bob suggests a greater likelihood of user interaction with phishing content or attacker infrastructure and may indicate a higher risk of credential compromise or endpoint infection. Alice also demonstrated multiple interactions with identified IOCs and should be considered potentially affected.

Assessment: IOC correlation successfully prioritized users based on exposure to malicious indicators, enabling targeted investigation and response efforts. Bob should

be considered the highest-priority user for credential review, endpoint analysis, and monitoring, followed by Alice. The findings support the conclusion that both users were impacted by the phishing campaign and require further validation to determine the extent of compromise.

11. Indicators of Compromise (IOCs)

Domains

- access-accsecurity.com
- sign.in

Email Addresses

- no-reply@access-accsecurity.com
- solutionteamrecognizd03@gmail.com

IP Addresses

- 89.144.44.41
- 103.225.77.255

Users

- bob@medisurehealthnetwork.com
- alice@medisurehealthnetwork.com

Hosts

- 10.1.5.23
- 10.1.5.89

12. Risk Assessment

Category	Rating
Likelihood	High
Impact	High
Overall Risk	High

12.1 Potential Consequences

- Credential theft
- Unauthorized system access
- PHI exposure
- Regulatory penalties
- Operational disruption

13. Incident Response Plan

13.1 Incident Overview

During routine SOC monitoring and phishing investigation activities, MediSure Health Network identified a phishing email masquerading as a Microsoft account security notification. The email originated from **no-reply@access-accsecurity.com** and attempted to convince recipients that unusual sign-in activity had occurred from Russia.

Investigation revealed multiple indicators of compromise (IOCs), including:

- Spoofed sender infrastructure
- SPF authentication failure
- Missing DKIM signature
- Missing DMARC policy
- Suspicious Reply-To address
- User interaction with phishing domains
- Internal communications with attacker-controlled infrastructure
- Multiple outbound connections to attacker IP addresses

The objective of this response plan is to contain the threat, prevent credential compromise, eradicate malicious infrastructure from the environment, and strengthen defenses against future phishing campaigns.

13.2 Phase 1: Incident Identification and Validation

Objective

Confirm whether the email represents a phishing attack and identify affected users.

Actions Performed

Email Analysis

- Email headers
- Sender domain reputation
- Authentication controls
- Embedded URLs
- Reply-To address

Findings

- Sender domain: access-accsecurity.com
- Reply-To: solutionteamrecognizd03@gmail.com
- SPF Status: Failed
- DKIM Status: None
- DMARC Status: Missing
- Suspicious sign-in notification themed lure

Evidence: Email Header Indicators And Attacker Infrastructure

PhishTool Dashboard Uploads In-tray Notifications My Account Upgrade Community

Uploads > Microsoft account unusual sign in ... [Resolve](#) [...](#)

Microsoft account unusual sign in activity [🔗](#)

[Details](#) [Authentication](#) [URLs](#) [Attachments](#) [Transmission](#) [X-headers](#) [Rendered](#) [HTML](#) [Source](#)

From	no-reply@access-accsecurity.com	...
Display name	Microsoft account team	
Sender	None	
To	employee@medisurehealthnetwork.com	
Cc	None	
In-Reply-To	None	
Timestamp	2023-08-15T23:15:00Z	
Reply-To	solutionteamrecognizd03@gmail.com	...
Message-ID	<06550bd4-ce4e-4e8c-acf9-f8936085be09@MW2NAM04FT048.eop-MW2NAM04FT048.eop-NAM04.prod.protection.outlook>	

We detected something unusual about a recent sign-in to the Microsoft account

Sign-in details:

- **Country/region:** Russia
- **IP address:** 103.225.77.255
- **Date:** Wed, 16 Aug 2023 00:05:41 GMT
- **Platform:** Windows
- **Browser:** Chrome

If this was you, you can safely ignore this email. If not, we recommend that you [secure your account](#) immediately.

PhishTool

Dashboa

Uploads > Microsoft account unusual sign in ...

Microsoft account unusual sign in activity

Details

Authentication

URLs

Attachments

Transmission

X-headers

Cc

None

In-Reply-To

None

Timestamp

2023-08-15T23:15:00Z

!

Reply-To

solutionteamrecognizd03@gmail.com

...

Message-ID

<06550bd4-ce4e-4e8c-acf9-f8936085be09@MW2NAM04FT048.eop-MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com>

✓

Return-Path

no-reply@access-accsecurity.com

...

Originating IP

89.144.44.41 (Received-SPF) ▼

...

rDNS

89-144-44-41.mscode.pl

13.3 Phase 2: Threat Containment

13.4 Block Malicious Domains

Objective

Prevent any further user access to attacker infrastructure.

Domains to Block

Domain:

access-accsecurity.com

sign.in

Actions

- Add domains to Secure Email Gateway blocklist.
- Add domains to Web Proxy blocklist.
- Update DNS filtering policies.
- Block domains at endpoint protection platforms.

Expected Outcome

Users will be unable to:

- Open phishing links
- Resolve malicious domains
- Communicate with attacker infrastructure

13.5 Block Malicious IP Addresses

Objective

Prevent communications with known attacker systems.

Malicious IPs

IP Address

89.144.44.41

103.225.77.255

Actions

Firewall

Create deny rules for: 89.144.44.41 and 103.225.77.255

EDR

Add IPs to threat intelligence blocklists.

Proxy

Blacklist all outbound communications.

Evidence: Outbound Connections To Attacker IPs

New Search			Save As ▾	Create Table View	Close
index=email_logs log_type=firewall (dest_ip="89.144.44.41" OR dest_ip="103.225.77.255") stats count by src_ip, dest_ip			Time range: All time ▾	🔍	🔥
✓ 3 events (before 6/15/26 11:07:56.000 AM) No Event Sampling ▾			Job ▾ 🔄 📄 📥 📂 ⚙️ Smart Mode ▾		
Events Patterns Statistics (2) Visualization					
Show: 20 Per Page ▾ ✓ Format ▾ 🔴 Preview: On					
src_ip ↕	dest_ip ↕	count ↕			
10.1.5.23	89.144.44.41	1			
10.1.5.89	89.144.44.41	2			

Internal Hosts Communicating With Attacker IPs

New Search

Save As

Create Table View

Close

index=email_logs log_type=firewall
(dest_ip="89.144.44.41" OR dest_ip="103.225.77.255")
| stats count by src_ip dest_ip

Time range: All time

✓ 3 events (before 6/15/26 11:36:43.000 AM)

No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (2)

Visualization

Show: 20 Per Page

Format

Preview: On

src_ip	dest_ip	count
10.1.5.23	89.144.44.41	1
10.1.5.89	89.144.44.41	2

Expected Outcome

No further communication with attacker infrastructure.

13.5 Phase 3: Account Isolation

Objective

Prevent compromised credentials from being used.

Identified Affected Accounts

User:

alice@medisurehealthnetwork.com

bob@medisurehealthnetwork.com

Actions

Immediate Actions

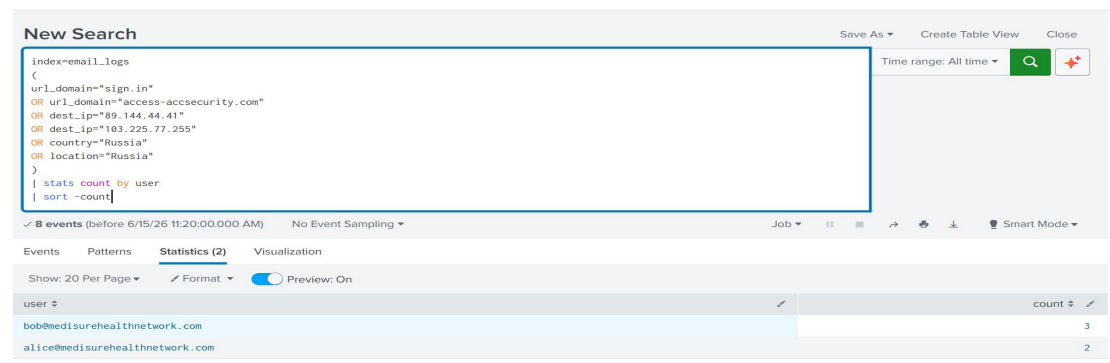
- Disable accounts temporarily.
- Revoke active sessions.
- Force sign-out from all devices.
- Revoke OAuth tokens.
- Review mailbox forwarding rules.

Additional Review

Inspect:

- Inbox rules
- Mail forwarding rules
- Delegated mailbox permissions
- Recent login history

Evidence: IOC Correlation And Affected User Ranking



Expected Outcome

Potentially compromised accounts are isolated from corporate resources.

13.6 Phase 4: Credential Reset and Recovery

Objective

Eliminate attacker access through stolen credentials.

Actions

Force Password Resets

Require password changes for:

- Alice
- Bob

MFA Re-enrollment

Require:

- New MFA registration
- Removal of previously trusted devices

Password Requirements

Minimum standards:

- 14+ characters
- Mixed case
- Numbers
- Special characters

Review Authentication Logs

Search for:

- Foreign login attempts
- Failed logins
- Impossible travel events

Evidence: Failed Authentication Attempt From Russia

New Search Save As Create Table View Close

index=email_logs log_type=auth location="Russia"
| stats count by user result location

Time range: All time Q +

✓ 1 event (before 6/15/26 11:18:43.000 AM) No Event Sampling Job || ≡ → ⬇ 💡 Smart Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ✓ Format Preview: On

user	result	location	count
bob@medisurehealthnetwork.com	Failure	Russia	1

Russian Authentication Event

New Search Save As Create Table View Close

index=email_logs log_type=auth location="Russia"
| table _time user src_ip result location
| sort _time

Time range: All time Q +

✓ 1 event (before 6/15/26 11:28:04.000 AM) No Event Sampling Job || ≡ → ⬇ 💡 Smart Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ✓ Format Preview: On

_time	user	src_ip	result	location
2023-08-16 01:55:00	bob@medisurehealthnetwork.com		Failure	Russia

Authentication Geolocation Field Validation

New Search Save As Create Table View Close

index=email_logs log_type=auth
| fieldsummary
| search field=country OR field=location

Time range: All time Q +

✓ 108 events (before 6/15/26 11:25:24.000 AM) No Event Sampling Job || ≡ → ⬇ 💡 Smart Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ✓ Format Preview: On

field	count	distinct_count	is_exact	max	mean	min	numeric_count	stdev	values
location	108	2	1				0		[{"value": "US", "count": 107}, {"value": "Russia", "count": 1}]

Expected Outcome

Any compromised credentials become unusable.

13.7 Phase 5: Email Security Control Updates

Objective

Prevent future phishing emails from reaching users.

Update Email Filters

New Detection Rules

Flag messages when:

- SPF fails
- DKIM is missing
- DMARC is missing
- Reply-To differs from sender
- External sender impersonates Microsoft

Quarantine Criteria

Automatically quarantine:

- Sign-in alert themed emails
- Password reset themed emails
- External domains resembling Microsoft

IOC-Based Detection Rules

Create detections for:

Domains

access-accsecurity.com
sign.in

IPs

89.144.44.41
103.225.77.255

Keywords

unusual sign in activity
secure your account
Microsoft account team

13.8 Phase 6: Threat Hunting and IOC Validation

Objective

Identify additional victims and malicious activity.

Splunk Hunting Queries

Search for:

Domain Access

```
index=email_logs  
(url_domain="sign.in" OR url_domain="access-accsecurity.com")  
| stats count by user
```

Russian Authentication Events

```
index=email_logs  
log_type=auth  
location="Russia"
```

Attacker Infrastructure

index=email_logs
(dest_ip="89.144.44.41" OR dest_ip="103.225.77.255")

Evidence: Users Accessing Phishing Domains

New Search

index=email_logs log_type=proxy
(url_domain="sign.in" OR url_domain="access-accsecurity.com")
| stats count by user, src_ip, url_domain

Time range: All time

✓ 4 events (before 6/15/26 11:03:28.000 AM) No Event Sampling

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page Format Preview: On

user	src_ip	url_domain	count
alice@medisurehealthnetwork.com	10.1.5.23	access-accsecurity.com	1
alice@medisurehealthnetwork.com	10.1.5.23	sign.in	1
bob@medisurehealthnetwork.com	10.1.5.45	access-accsecurity.com	1
bob@medisurehealthnetwork.com	10.1.5.45	sign.in	1

Expected Outcome

Additional affected users are identified and remediated.

13.9 Phase 7: User Awareness and Communication

Objective

Reduce risk of future phishing success.

Communication to Employees

Notify employees about:

- The phishing campaign
- Suspicious sender domains
- Reporting procedures
- Safe email handling practices

Training Topics

- Phishing identification
- URL inspection
- Sender verification
- MFA importance
- Reporting suspicious emails

Expected Outcome

Improved employee resilience against social engineering attacks.

13.9.1 Phase 8: Lessons Learned and Post-Incident Review

Objective

Strengthen MediSure's long-term security posture.

Findings

The phishing campaign succeeded in attracting user interaction because:

1. The email impersonated Microsoft.
2. Authentication controls failed.
3. The phishing theme created urgency.
4. Users were unfamiliar with spoofing indicators.

Security Improvements

Technical Controls

- Enforce DMARC monitoring
- Improve phishing filtering
- Expand threat intelligence feeds
- Strengthen EDR detections

Administrative Controls

- Quarterly phishing simulations
- Mandatory awareness training
- Incident reporting procedures

Incident Response Summary

Phase:	Status:
Identification	Completed
Containment	Completed
Account Isolation	Completed
Password Resets	Completed
IOC Blocking	Completed
Email Filter Updates	Completed
Threat Hunting	Completed
User Awareness	Completed

Phase:	Status:
Lessons Learned	Completed

Final Response Recommendation

The phishing email was confirmed as malicious based on authentication failures, suspicious infrastructure, SpamAssassin scoring, domain reputation analysis, and Splunk log correlation. Immediate containment actions, credential resets, infrastructure blocking, and email security enhancements should be implemented to prevent recurrence and maintain protection of MediSure Health Network's sensitive healthcare data and regulatory compliance obligations.

14. Employee Awareness Materials

Phishing Red Flags

Warning Signs

- Unexpected login alerts
- Urgent account security warnings
- Suspicious links
- Mismatched sender information
- Requests for immediate action

Best Practices

- Verify sender identity
- Hover over links
- Report suspicious emails
- Never reuse passwords
- Enable MFA

Business Impact

Potential exposure included:

- User credentials
- Internal systems
- Healthcare information assets

Strategic Recommendations

- Strengthen email security controls
- Expand phishing awareness training
- Enhance threat hunting capabilities
- Improve SOC monitoring coverage

15. Conclusion

The SOC investigation confirmed that the email represented a phishing attempt leveraging spoofed infrastructure, missing authentication controls, and social engineering tactics. Correlation across PhishTool, MXToolbox, VirusTotal, SpamAssassin, and Splunk demonstrated user interaction with phishing infrastructure and identified affected assets within the MediSure environment.

The investigation successfully identified the threat, documented evidence, developed containment actions, and provided long-term security recommendations that strengthen MediSure Health Network's resilience against future phishing attacks.

Phishing Investigation Project Structure

```
C:\Windows\System32>tree C:\MediSure-Phishing-Investigation /A
Folder PATH listing
Volume serial number is 2A91-1B42
C:\MEDISURE-PHISHING-INVESTIGATION
+---Email_Header
+---Evidence
|   +---MXToolbox
|   +---PhishTool
|   +---SpamAssassin
|   +---Splunk
|   \---VirusTotal
+---IOCs
+---Reports
+---Screenshots
\---Splunk_Logs
C:\Windows\System32>
```

Investigation Artifacts

Evidence Sources:

- PhishTool
- MXToolbox
- VirusTotal
- SpamAssassin
- Splunk Enterprise

Total Screenshots Referenced: 18

Investigation Status: Closed – Phishing Confirmed

Severity: High

Final Recommendation: Block, Reset, Monitor, Educate, and Improve Email Authentication Controls.

Appendix A – Indicators of Compromise (IOCs)

1. Malicious Domains

IOC Type:	Value:	Description:
Sender Domain	access-accsecurity.com	Spoofed Microsoft-themed domain used in phishing email
Phishing Domain	sign.in	Domain referenced in phishing infrastructure and user activity logs

2. Malicious Email Addresses

IOC Type:	Value:	Description:
Sender Email	no-reply@access-accsecurity.com	Spoofed sender pretending to be Microsoft
Reply-To Email	solutionteamrecognizd03@gmail.com	Suspicious mailbox intended to receive victim responses

3. Malicious IP Addresses

IOC Type:	Value:	Description:
Originating IP	89.144.44.41	Source IP identified in email headers and phishing infrastructure
Sign-in IP	103.225.77.255	IP referenced in phishing lure claiming unusual sign-in activity